
Big Data Analytics
Pitchaya Wiratchotisatian



https://thebaldbrothers.com/50-best-hobbies-for-men/

SC663402 Data Warehouse and Big Data Analytics

*SC 663 402 **คลังข้อมูลและการวิเคราะห์ข้อมูลขนาดใหญ่** 3(2-2-5)

Data Warehouse and Big Data Analytics

เงื่อนไขของรายวิชา: SC 310 003#

แนวคิดของคลังข้อมูล มาร์ทข้อมูล ทะเลสาบข้อมูล คิวบ์ข้อมูลขั้นแนะนำ โครงสร้างรูปแบบต่าง ๆ และการประมวลผลเชิงวิเคราะห์ออนไลน์ในฐานข้อมูลแบบพหุมิติ การทำให้เกิดคลังข้อมูล เทคโนโลยีคลาวด์สำหรับการประมวลผลข้อมูลขนาดใหญ่ ฮาดูป สปาร์ค ภาษาไฮว์และพิก ฐานข้อมูลแบบโนเอสคิวแอล

Concept of data warehouse and data mart, data lake, introduction to data cubes, various types of schemas and on-line analytical processing (OLAP) in multidimensional database, data warehouse implementation, cloud technology for big data processing: Hadoop, Spark, Hive/Pig, noSQL database

- Cloud technology for big data processing:
  Hadoop, Spark, Hive/Pig, noSQL database

Outline

- What is Big Data?
- The process of Big Data Analytics
  - Hadoop, Spark, Hive/Pig, noSQL database
- Text Analytics
- Deep Learning
- Cloud technology

Course Outline

ช่องทางการสื่อสาร

Google Classroom

อุปกรณ์การเรียน: มือถือ, คอมพิวเตอร์พกพา (ใครไม่มี?)

กติกาการเรียน

- Academic dishonesty
  - Plagiarism
- Group project

Data Analytics: Great!

Experiments, observations, and numerical simulations in many areas of science and business are currently generating terabytes of data, and in some cases are on the verge of generating petabytes and beyond. Analyses of the information contained in these data sets have already led to major breakthroughs in fields ranging from genomics to astronomy and high-energy physics and to the development of new information-based industries.

- Frontiers in Massive Data Analysis, National Research Council of the National Academies

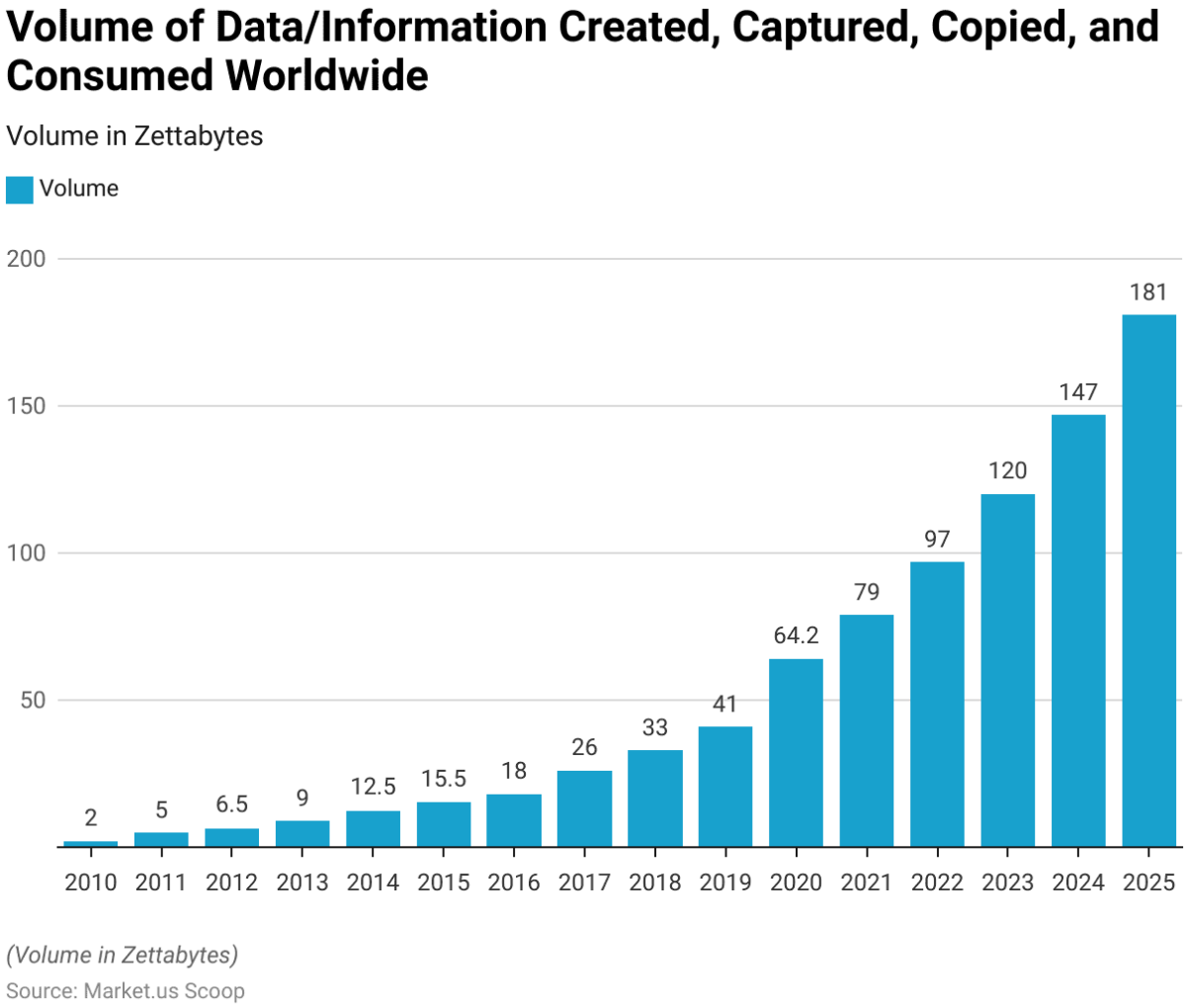
Data Analytics: Caution!

Statistical rigor is necessary to justify the inferential leap from data to knowledge, and many difficulties arise in attempting to bring statistical principles to bear on massive data. Overlooking this foundation may yield results that are not useful at best, or harmful at worst. In any discussion of massive data and inference, it is essential to be aware that it is quite possible to turn data into something resembling knowledge when actually it is not. Moreover, it can be quite difficult to know that this has happened.

- Frontiers in Massive Data Analysis, National Research Council of the National Academies

How Data Is Measured

- A single byte is composed of 8 bits (of 0's or 1's)
- Larger units for measuring data include kilobytes (KB), megabytes (MB), gigabytes (GB), terabytes (TB), petabytes (PB), exabytes (EB), and zettabytes (ZB)
  - Each unit in the hierarchy is 1,000 times larger than the previous one
  - e.g., 1 TB = 1,000 GB
- https://www.youtube.com/watch?v=x4jPg0Nw2BI



- **402.74 million** terabytes of data is generated every day
- **90%** of the world's data was created in the last two years
- By 2025, over **80%** of organizations anticipate managing zettabytes of data, yet 36% admit they won't be able to handle it all
- The big data market is supposed to grow to **$655 billion** by 2029

https://www.g2.com/articles/big-data-statistics

DATA CREATED EVERY DAY

181,000,000,000,000,000,000,000

bytes of data predicted to be **created by 2025**.

16,000,000

texts sent **every minute**.

328,770,000,000,000,000,000

bytes of data created daily.

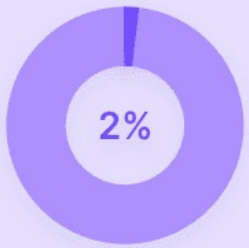
361,000,000,000

emails every day in 2024.

The amount of **data generated worldwide** has soared from 2 ZB in 2010 to a whopping **64.2 ZB in 2020**

which is more than the number of detectable stars in the cosmos.





A mere **2% of the data created and consumed** in 2020 **was saved and stored** up to 2021.



Storage capacity is forecast to increase **from a current installed base of 6.7 ZB** in 2020 at a compound **annual growth rate (CAGR) of 19.2%** to 2025.

https://spacelift.io/blog/how-much-data-is-generated-every-day

spacelift

Key Sources of New Data

- AI-Generated Content
- Social Media and User-Generated Content: Social media platforms are major contributors, with billions of daily uploads and interactions on platforms like TikTok, YouTube, and Instagram, adding to a growing pool of both structured and unstructured data
- IoT Devices and Smart Technology: The Internet of Things (IoT) has turned devices such as smart home appliances, wearable health monitors, and industrial sensors into constant real-time data generators

Key Sources of New Data

- Enterprise and Transactional Data: Businesses create large amounts of data through operational systems like ERP platforms, customer relationship management tools, and supply chain management software with every transaction
- Scientific Research and Big Data: Fields like genomics, climate science, and particle physics generate huge datasets for simulations, experiments, and analysis using tools like telescopes and genome sequencers
- Cloud Computing and the Shift to Digital Storage: Cloud systems enable the easy storage, access, and processing of massive amounts of data, lifting storage limitations and supporting real-time data processing and analytics

Key Sources of New Data

- eCommerce: eCommerce platforms generate continuous transactional and behavioral data from every search, click, purchase, and review, which powers recommendation engines and personalized marketing
- Streaming: Video, music, and gaming platforms create data with every user interaction, with the push for higher-quality content like 4K and 8K video further increasing the data load
- Digital Transactions: Digital payment systems like mobile wallets and cryptocurrencies generate a trail of financial data with every transaction, with blockchain technology storing records across decentralized networks

https://rivery.io/blog/big-data-statistics-how-much-data-is-there-in-the-world/

5.07 billion people are using the Internet (Source: CNBC TV18)

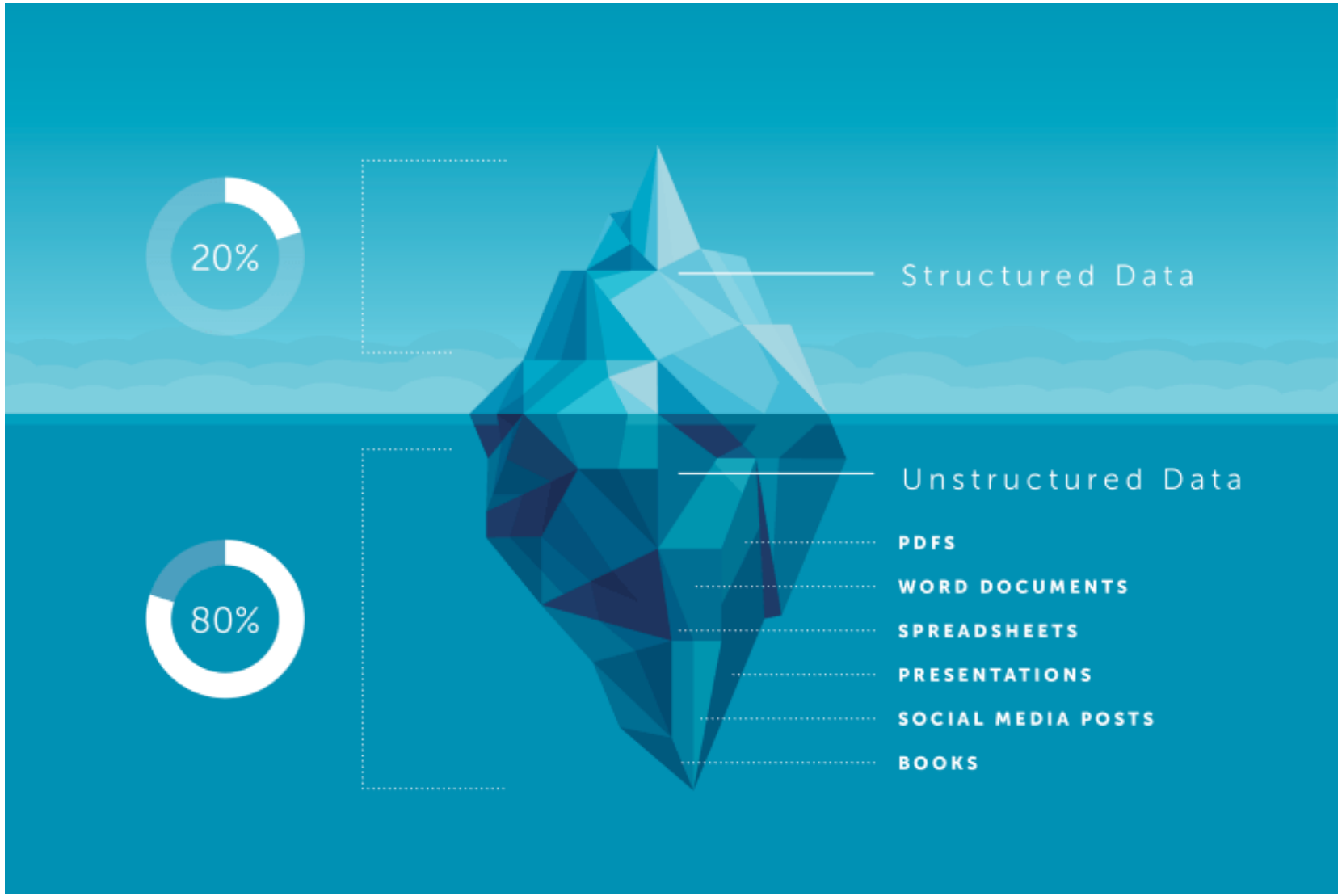


Data is a new gold!



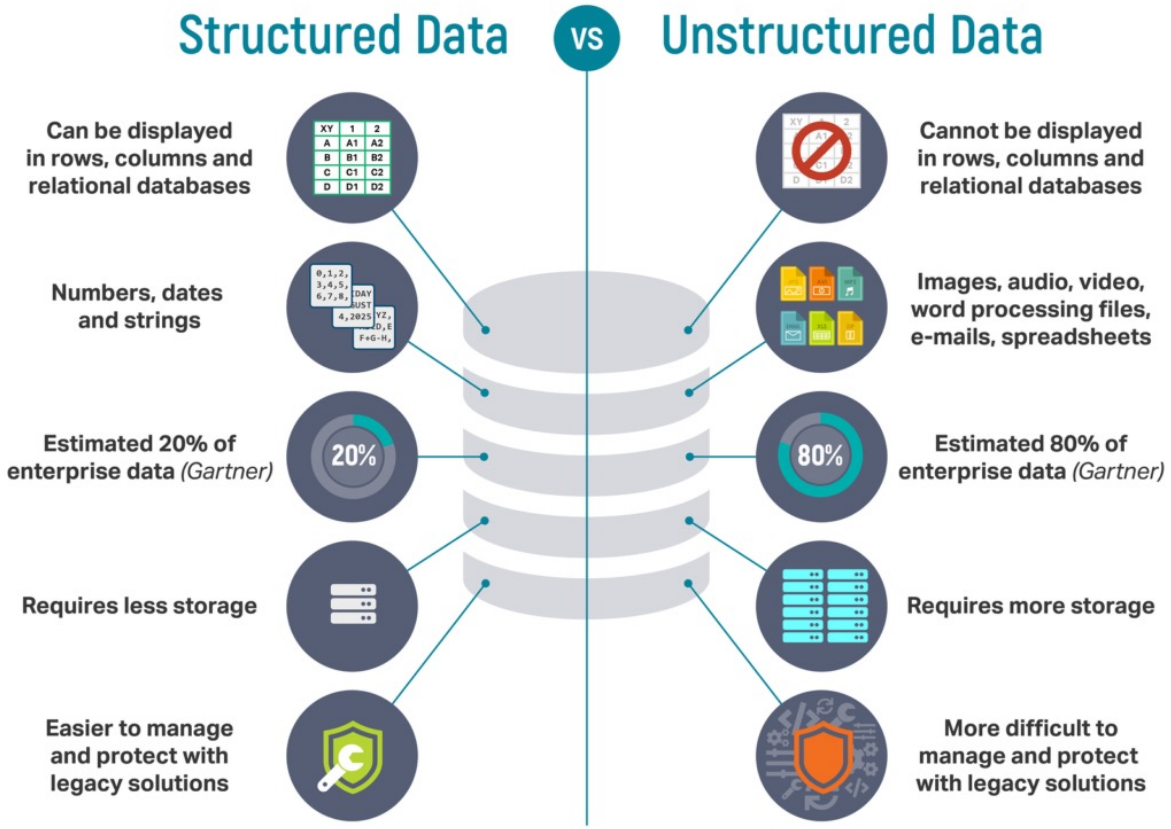
https://www.ceotodaymagazine.com/2018/04/is-data-the-new-gold/

Unstructured Data



https://www.igneous.io/blog/structured-data-vs-unstructured-data

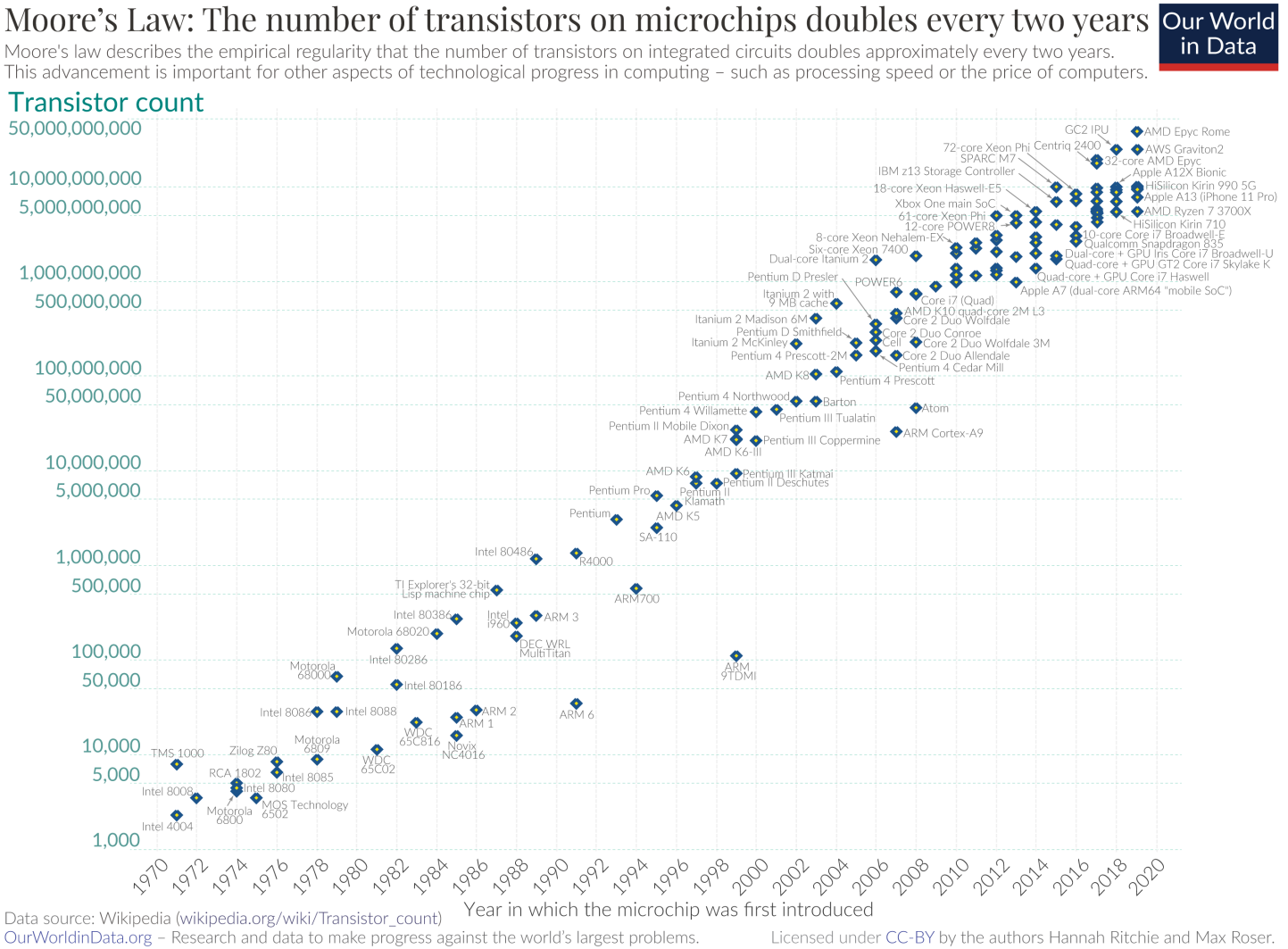
Unstructured Data



https://www.igneous.io/blog/structured-data-vs-unstructured-data

Why Big Data?

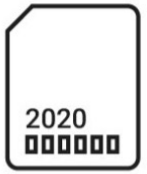
- Computing has gotten really cheap…
- Gordon Moore predicted that the number of transistors in silicon chips would double every 2 years
- Computers would increase in power and decrease in relative cost at an exponential pace
- Moore’s Law held up for more than 50 years, until 2 years ago



A semi-log plot of transistor counts for microprocessors against dates of introduction, nearly doubling every two years.
(https://en.wikipedia.org/wiki/Moore%27s_law#/media/File:Moore's_Law_Transistor_Count_1970-2020.png)

Big Data Consists of 4 V's: ขนาดใหญ่มาก!

40 ZETTABYTES
of data will be created by 2020, an increase of 300 times from 2005



6 BILLION PEOPLE
have cell phones
WORLD POPULATION: 7 BILLION



Volume
SCALE OF DATA

2.5 QUINTILLION BYTES
of data are created each day

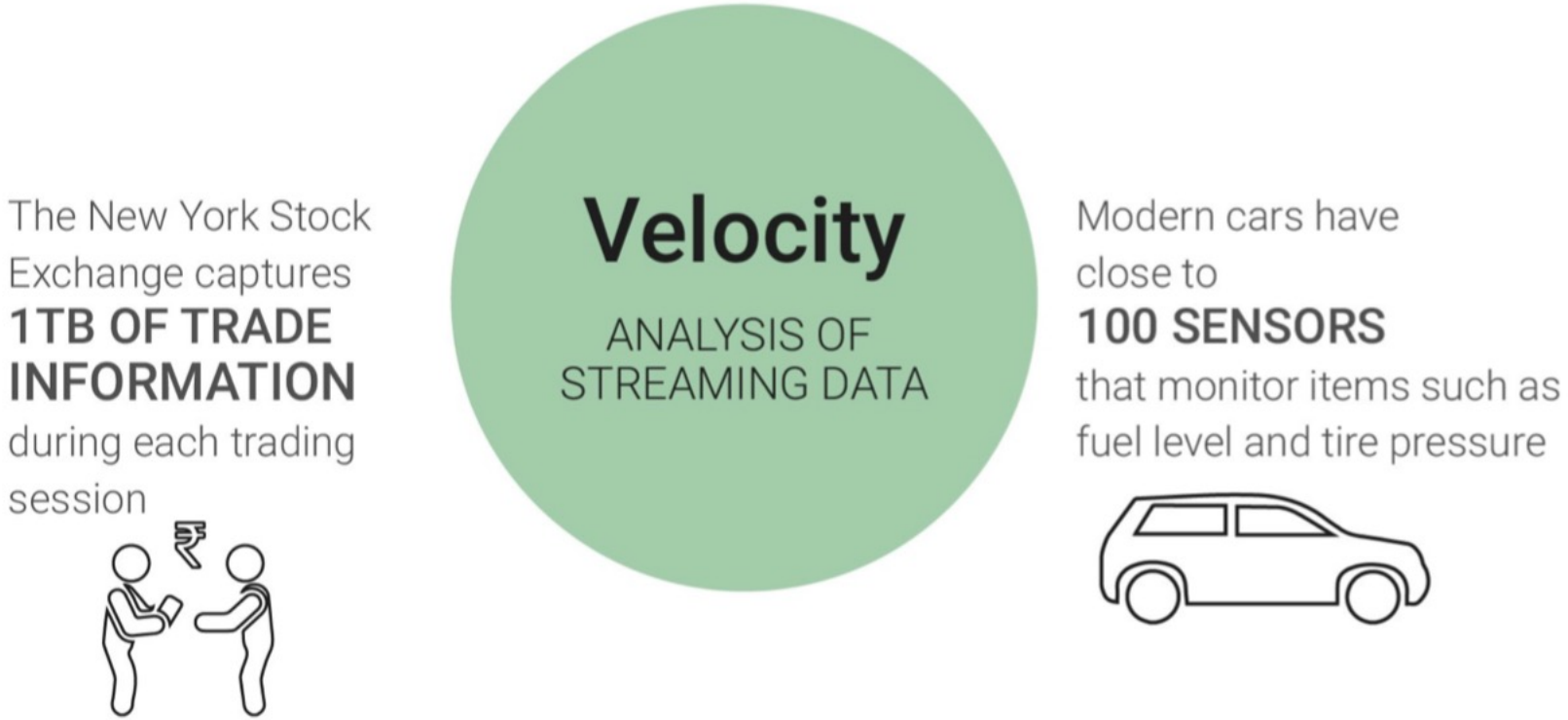


Most companies in the U.S. have at least
100 TERABYTES
of data stored



https://twitter.com/iauro/status/889725873707827200?s=20&t=_dZZtelc1Ut8UhZrQlRrHA

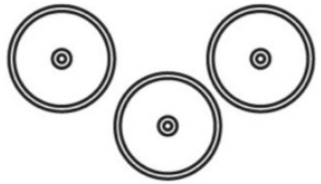
Big Data Consists of 4 V's: รวดเร็ว สดใหม่



https://twitter.com/iauro/status/889725873707827200?s=20&t=_dZZteIc1Ut8UhZrQlRrHA

Big Data Consists of 4 V's: หลากหลาย

As of 2011, the global size of data in healthcare was estimated to be
150 EXABYTES



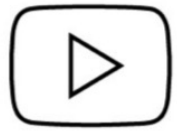
30 BILLION PIECES OF CONTENT
are shared on facebook every month



Variety

DIFFERENT FORMS OF DATA

4 BILLION + HOURS OF VIDEO
are watched on You Tube each month

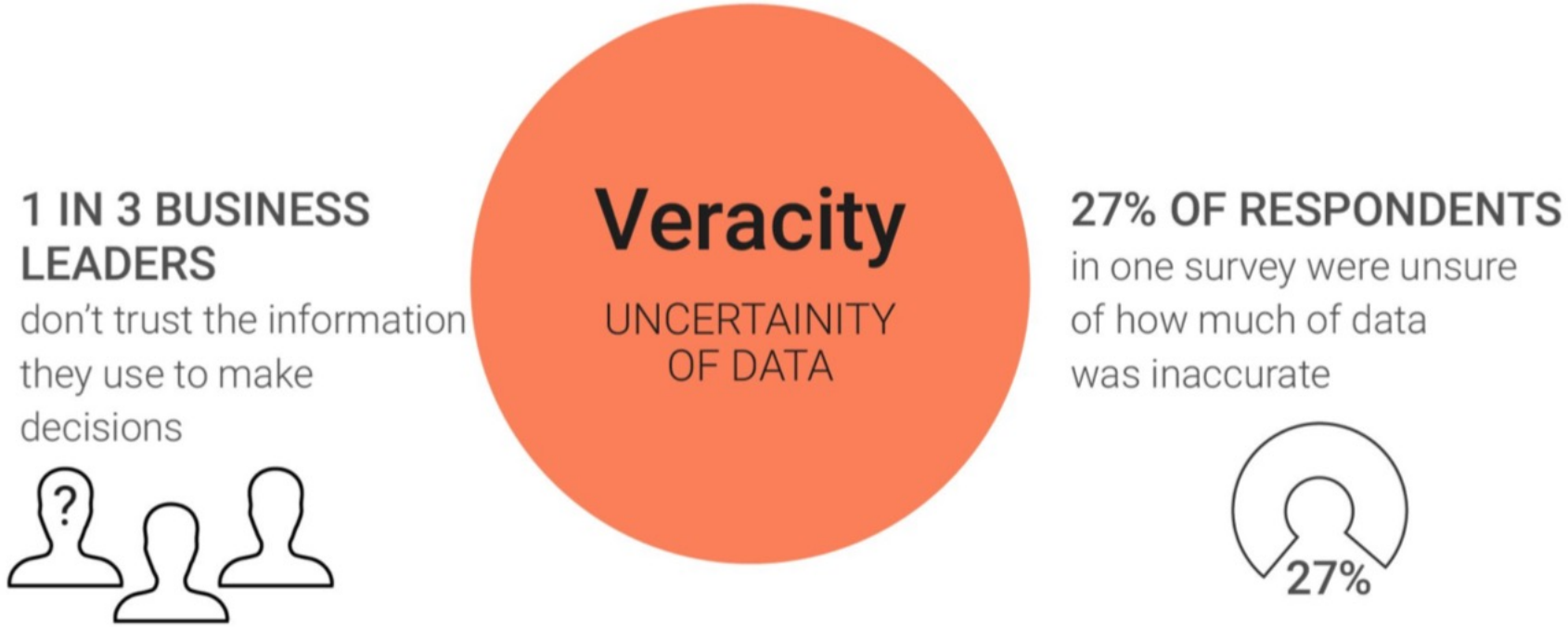


4 MILLION TWEETS
are sent per day by about 200 million monthly active users



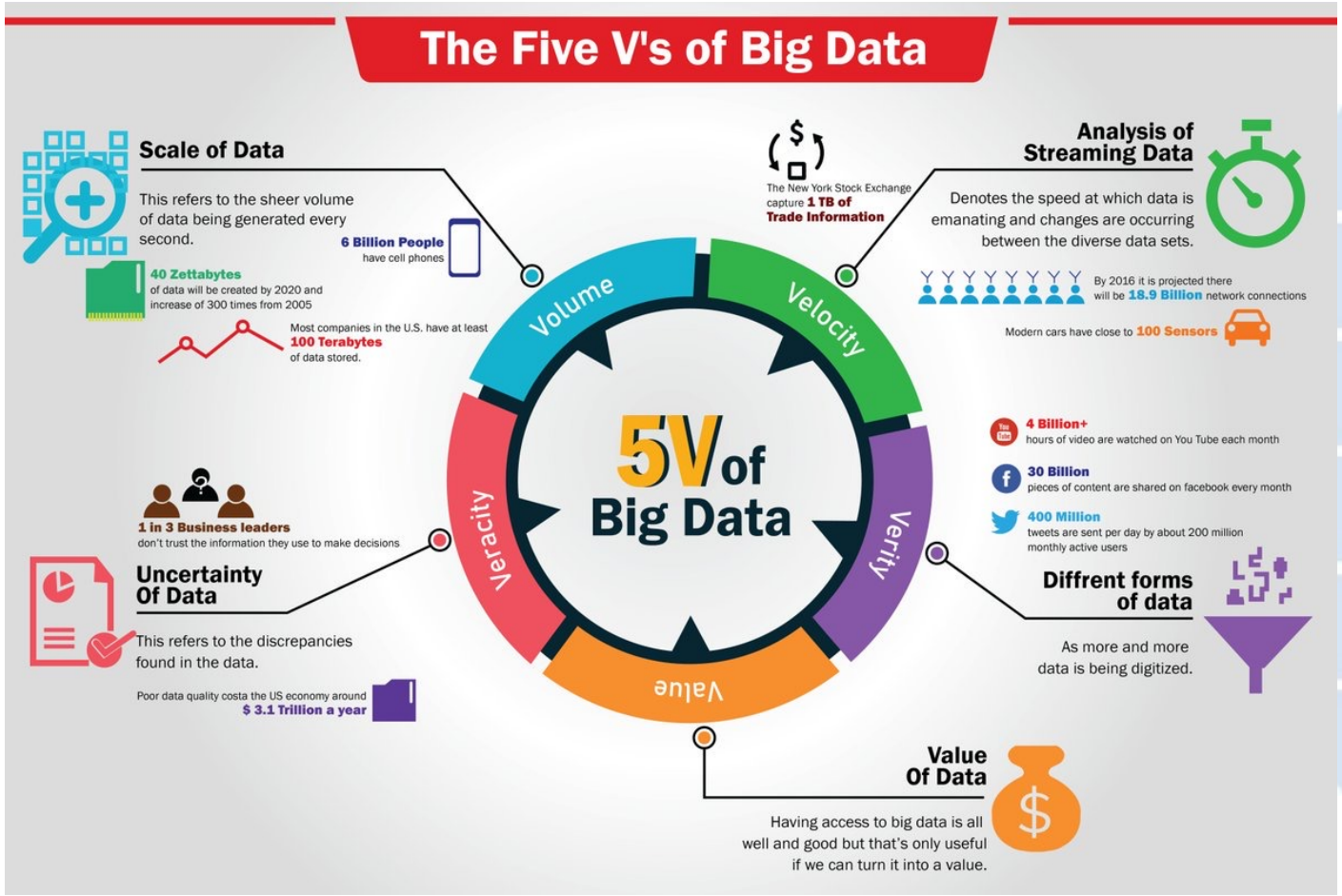
https://twitter.com/iauro/status/889725873707827200?s=20&t=_dZZtelc1Ut8UhZrQlRrHA

Big Data Consists of 4 V's: ความน่าเชื่อถือ



https://twitter.com/iauro/status/889725873707827200?s=20&t=_dZZtelc1Ut8UhZrQlRrHA

Another V of Big Data

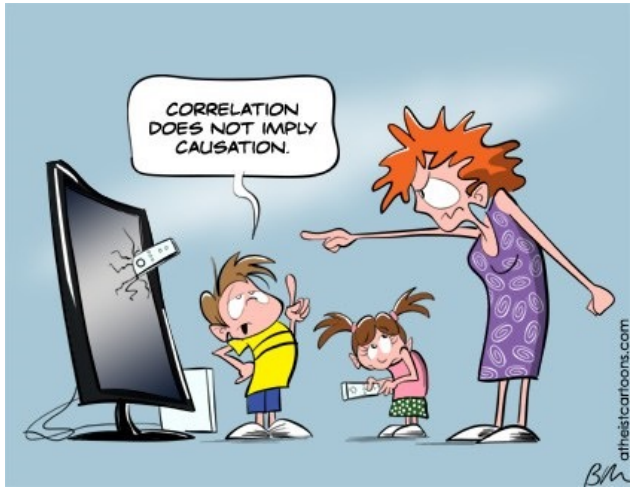


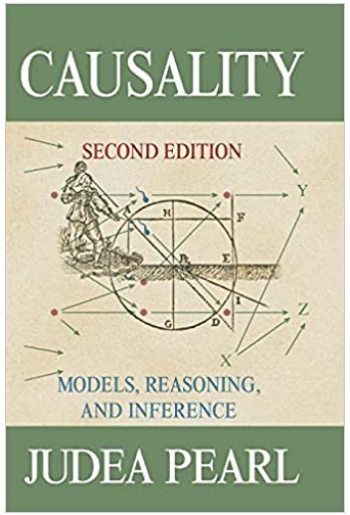
Value	Clinically relevant data Longitudinal studies
Volume	High-throughput technologies Continuous monitoring of vital signs
Velocity	High-speed processing for fast clinical decision support Increasing data generation rate by the health infrastructure
Variety	Heterogeneous and unstructured data sources Differences in frequencies and taxonomies
Veracity	Data quality is unreliable Data coming from uncontrolled environments
Variability	Seasonal health effects and disease evolution Non-deterministic models of illness and health

https://twitter.com/execlvlsearch/status/1174714224410599424

Getting Knowledge from Data

Are we searching for causality?



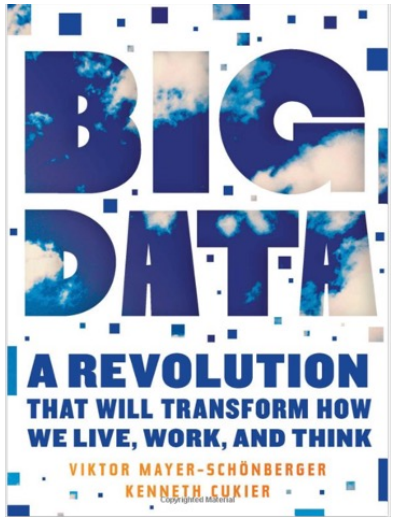


“Causation: The relation between mosquitoes and mosquito bites. Easily understood by both parties but never satisfactorily defined by philosophers and scientists.”

- http://freshspectrum.com/causation/ Michael Scriven, Evaluation Thesaurus, 1991

Getting Knowledge from Data

Are we searching for causality?



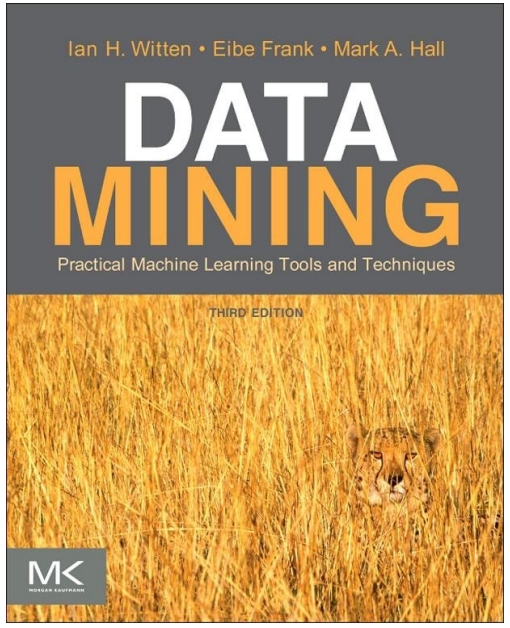
“Most strikingly, society will need to shed some of its obsession for causality in exchange for simple correlations: not knowing why but only what.”

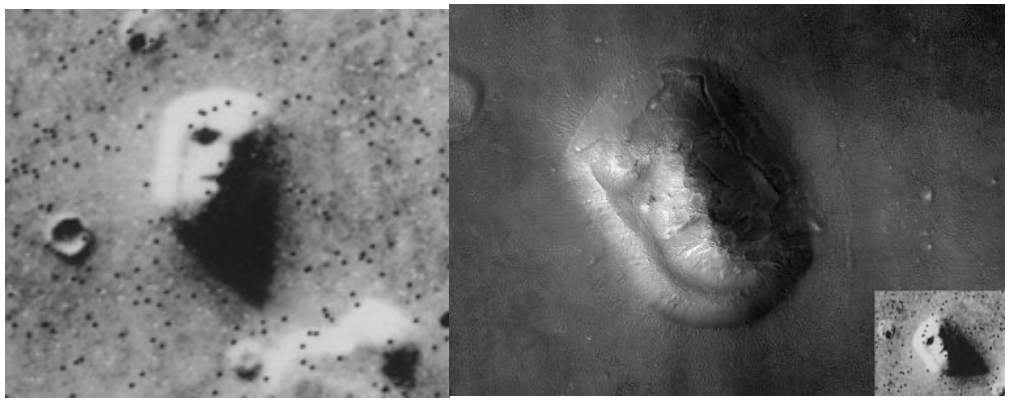
- Big Data: A Revolution that Transform How We Live, Work, and Think, Viktor Mayer-Schönberger and Kenneth Cukier.

Getting Knowledge from Data

Are humans or computers better at finding patterns or making predictions from data?

- Human beings are really good at pattern detection. Perhaps a bit too good!

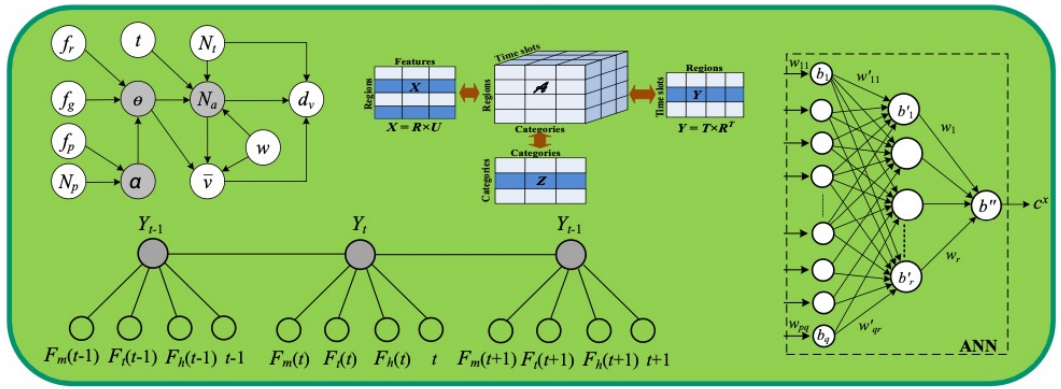




(1) Face on Mars appeared due to data errors (salt-and-pepper noise) giving the image a speckled appearance
(2) Mars Reconnaissance Orbiter image by its HiRISE camera of the "Face on Mars" (2007)
https://en.wikipedia.org/wiki/Cydonia_(Mars)

Data Scientist

Models /Algorithms/AI



Problems









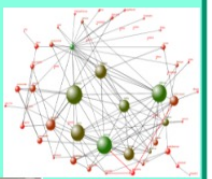


Data









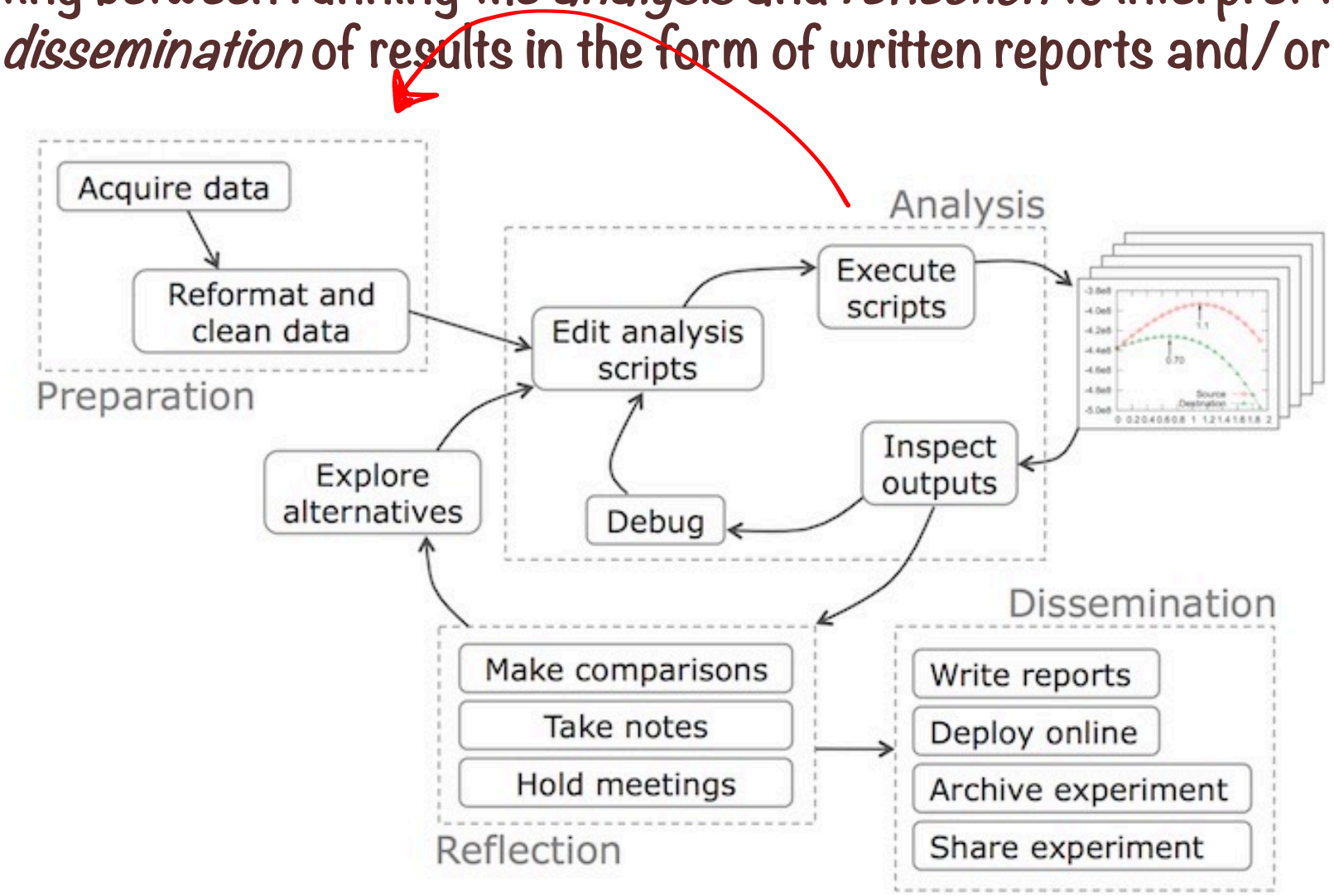






Typical data science workflow. There are four main phases, shown in the dotted-line boxes:

- *preparation* of the data,
- alternating between running the *analysis* and *reflection* to interpret the outputs, and
- finally, *dissemination* of results in the form of written reports and/or executable code



https://cacm.acm.org/blogs/blog-cacm/169199-data-science-workflow-overview-and-challenges/fulltext

Learning Data Analytics

Besides knowledge, there are three keys that make a great (Big) Data Analytics learning experience

1. Working on a project actually using data!

Learning Data Analytics

Besides knowledge, there are three keys that make a great (Big) Data Analytics learning experience

1. Working on a project actually using data!
2. Working on a project actually using data!!

Learning Data Analytics

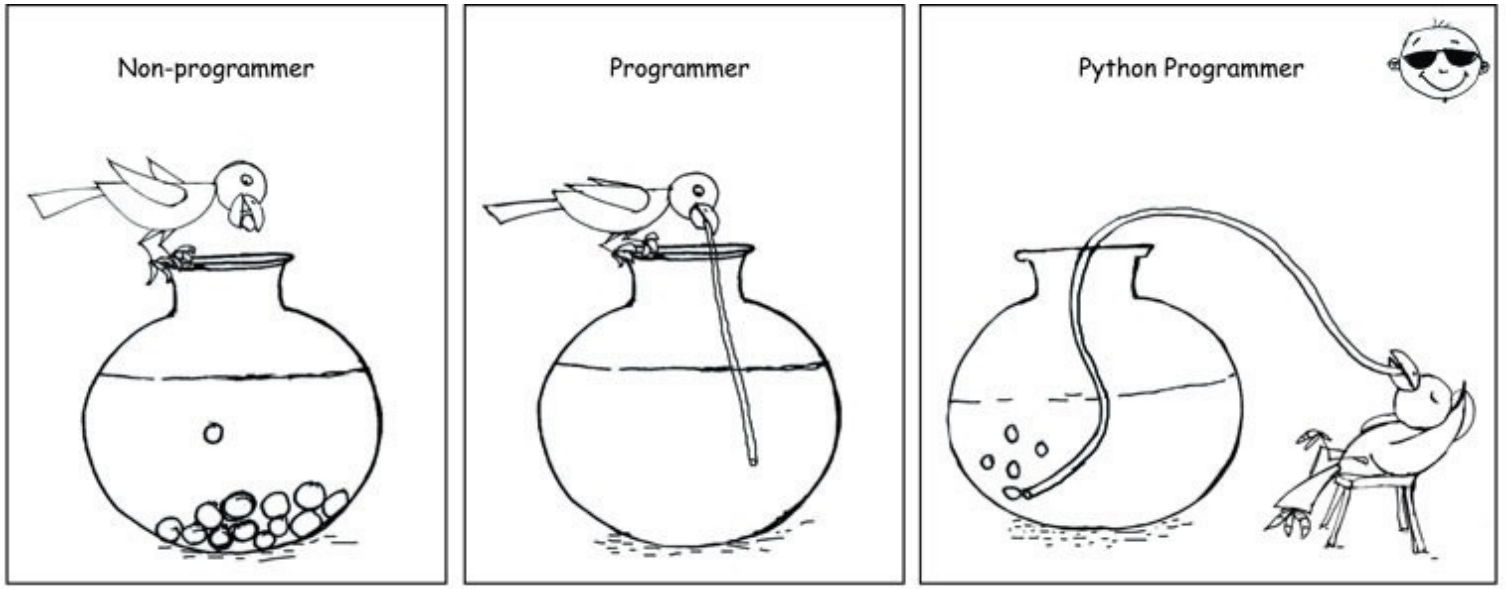
Besides knowledge, there are three keys that make a great (Big) Data Analytics learning experience

1. Working on a project actually using data!
2. Working on a project actually using data!!
3. Working on a project actually using data!!!



Python 🐍

- Great and free programming model
- Access to other languages like C, C++, JAVA, etc.
- Lots of libraries: Numpy, scipy, pandas, pycuda, mpi4py, etc.



http://lpycot.appspot.com/

Install Python

- The default Python prompt through command line or scripts
- The IPython interpretor through command line or scripts
- The Jupyter notebook, a beautiful integrated development environment (IDE)
- Other ways such as Spyder, included with Anacoda
- Google Colaboratory (aka Google Colab), which is an online notebook environment that runs on Google Cloud
  - The user interface is similar to a Jupyter notebook
  - It supports GPU and TPU for machine learning models with limited space (15GB) and time (12 hrs/day)
  - Tutorial: https://pub.towardsai.net/google-colab-tutorial-for-beginners-834595494d44

Note

Tutorials

- Python
- Jupyter Notebook
- Google Colab